Economic Complexity 2026 — Written Exam 1

Oct 20, 2026

Instructions

- Time: 30 minutes. Total: 15 points.
- Answer all questions in Part A and one question in Part B.
- Show your calculations. A pocket calculator is allowed.
- Where relevant, refer to concepts and authors from the readings and lecture slides (weeks 1–5).

Part A — Short answers (answer all; 4 × 2.5 = 10 points; ~20 minutes)

A1. Same diversity, different complexity (2.5 points).

Countries X and Y both have revealed comparative advantage (RCA ≥ 1) in exactly 20 products. Country X's products are exported by only a few other countries. Country Y's products are exported by most countries in the world.

1. Which country is likely more complex? Explain using the capabilities logic of Hidalgo and Hausmann (2009). (1 point)
2. Why does the Economic Complexity Index need both diversity and ubiquity, rather than diversity alone? (1.5 points)

A2. Measuring relatedness (2.5 points).

1. A statistical office groups products into categories such as "capital-intensive manufacturing" or "labor-intensive agriculture". Why does the product space measure proximity from co-exports instead of using such categories? (1 point)
2. Region R is specialized in products a, b and c. Product d is close to a, b and c in the product space; product e is close to none of them. Which product is R more likely to enter in the next ten years, and what does the principle of relatedness assume about why? (1.5 points)

A3. The geography of innovation (2.5 points).

Audretsch and Feldman (1996) compare how concentrated innovation and production are across US states, using 4,200 manufacturing innovations from 1982. In the computer industry, 41.7% of all innovations came from California alone.

1. A critic argues: "Computer innovation is concentrated in California only because computer production is concentrated there." How does the paper's design respond to this critique? What result would have contradicted the knowledge-spillover explanation? (1 point)
2. Today, much research is shared online and teams collaborate remotely. Using the paper's argument, predict whether innovation in industries that rely on university research should become more or less geographically concentrated. Justify your prediction and name one piece of evidence that could support or contradict it. (1.5 points)

A4. Revealed skill relatedness (2.5 points).

In a labor-flow dataset, 20,000 workers switch industries in total. Industry i sends 1,000 workers to other industries, and industry j receives 2,000. We observe 200 workers moving from i to j.

1. Following Neffke, Otto and Weyh (2017), compute the expected flow from i to j, the ratio R of observed to expected flows, and its normalized value $(R - 1)/(R + 1)$. (1.5 points)
2. Why compare observed flows to expected flows instead of using raw flow counts? (0.5 points)
3. Give one reason the authors prefer labor flows over co-location as a measure of relatedness. (0.5 points)

Part B — Applied reasoning (choose 1; 5 points; ~10 minutes)

B1. Reading a relatedness regression.

A researcher studies which industries European regions enter between 2010 and 2020. Entry = 1 if a region's RCA in an industry rises from below 0.5 in 2010 to at least 1 in 2020, and 0 otherwise. The sample contains region–industry pairs with RCA below 0.5 in 2010. On average, 5% of these pairs enter. The table shows a hypothetical linear probability model. The three density variables measure how many related industries the region already specializes in, using three different relatedness measures. They are standardized (mean 0, standard deviation 1).

Variable	Coefficient
Relatedness density: skill relatedness (labor flows)	+0.045***
Relatedness density: co-location	+0.012*
Relatedness density: input–output links	+0.004
Log regional employment	+0.010**
Region and industry fixed effects	Yes
N / R^2	24,000 / 0.11

1. Explain what these results tell us about how regions diversify. Interpret the main coefficient and connect the comparison of the three relatedness measures to the course readings. (2 points)
2. A regional development agency wants to use this model to decide which industries to support. Write a short critical assessment: how far can the agency trust and use these results? Consider how entry and the controls are defined, whether the effects are causal, and what the model cannot tell the agency. (3 points)

B2. "Enter the most related and most complex products."

A minister of economy reads the relatedness–complexity diagram and proposes to subsidize only the products that are both highly related to the current export basket and highly complex. Evaluate this proposal with three arguments from the course. Use at least one argument from weeks 1–2 (relatedness and complexity) and one from week 5 (skills and labor mobility). End with a one-sentence alternative strategy. (5 points)

B3. Labor-flow networks and regional resilience.

Regions X and Y have the same employment and the same number of industries. In X, most significant inter-industry labor flows pass through one large automotive industry. In Y, labor flows connect many industries without one dominant hub. A recession hits both regions, and the automotive industry in X announces large layoffs.

1. Argue which region is likely to recover better and why. Use the structure of the two labor-flow networks, what happens to laid-off workers, and the idea of skill relatedness. You may sketch the two networks. (3 points)
2. Propose how you would test your argument empirically across many regions: what data you need, how you would measure the network structure, and which outcome you would compare. (2 points)